



Carbon credit, carbon trading and carbon footprint

Carbon credit

A carbon credit is a permit that allows the emission of a certain amount of carbon dioxide (CO₂).

Parties with commitments under the Kyoto Protocol (Annex B Parties) have accepted targets for limiting or reducing emissions.

These targets are expressed as levels of allowed emissions, or assigned amounts, at over the 2008–2012 commitment period.

The allowed emissions are divided into assigned amount units (AAUs).

A company can reduce carbon emission by following new eco-friendly technologies which in turn may let them earn carbon credits



INTERNATIONAL EMISSIONS TRADING(IET) - Carbon Trading

- Emissions trading, as set out in Article 17 of the Kyoto Protocol
- Countries with commitments under the Kyoto Protocol can acquire emission units from other countries with commitments under the Protocol and use them towards meeting a part of their targets.
- An international transaction log, a software-based accounting system, ensures secure transfer of emission reduction units between countries



Carbon footprint

- **Carbon footprint**, amount of carbon dioxide (CO₂) emissions associated with all the activities of a person or other entity (e.g., building, corporation, country, etc.).
- It includes direct emissions, such as those that result from fossil-fuel combustion in manufacturing, heating, and transportation, as well as emissions required to produce the electricity associated with goods and services consumed.
- In addition, the carbon footprint concept also often includes the emissions of other greenhouse gases, such as methane, nitrous oxide, or chlorofluorocarbons



Carbon footprint

- The average carbon footprint for a person in the United States is 16 tons, one of the highest rates in the world.
- Globally, the average is closer to 4 tons.
- To have the best chance of avoiding a 2°C rise in global temperatures, the average global carbon footprint per year needs to drop under 2 tons by 2050.



Reducing Carbon Footprint

Energy efficiency at home – All appliances that are not being used must be switched off immediately. And all electrical outlets not in use must also be switched off. Hot-water geysers should be switched off for the entire day and only turned on when needed. These are simple, yet practical lifestyle habits which are easy to adopt.

Buy renewable energy – It is quite possible to power your own home with environmentally sustainable alternatives of energy production without compromising your lifestyle and waiting for national grids to be connected via green energy supply sources. For instance, technology is now available for you to install your own solar power panels.

Recycle and re-use – Vegetable produce can be converted into compost (or manure) for gardens, even vegetable gardens. Instead of buying more food containers, plastic containers sourced from the supermarket can be refashioned as ideal kitchen utensils. Also, where plastic waste is no longer required, seek out recycling depots rather than relying on your supplied garbage disposal units.

Plant a Tree – One of the best way to give it back to the environment is to plant trees. Plants absorb CO₂ and release oxygen that is then used by humans and animals. According to the Urban Forestry Network, a single young tree absorbs 13 pounds of carbon dioxide each year.

Buy local – Adding to the above remark, buying local, organic produce effectively counters mass produced agricultural outcomes. There is a dramatic reduction in the amount of plastic being used to package products and fuel usage during long road transits is also reduced.

What is Carbon Footprinting?

“Commonly used to describe the total amount of CO₂ and other greenhouse gas (GHG) emissions attributable to an organisation, project or product.”



Sources of Carbon Emissions from your Organisation

Your Suppliers

- *Materials, goods and services,*
- *Capital goods,*

Embodied carbon

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Upstream



Your Business

- *Fuel and energy in company facilities*
- *Vehicles and plant.*

Operational Carbon

Company



Your Client

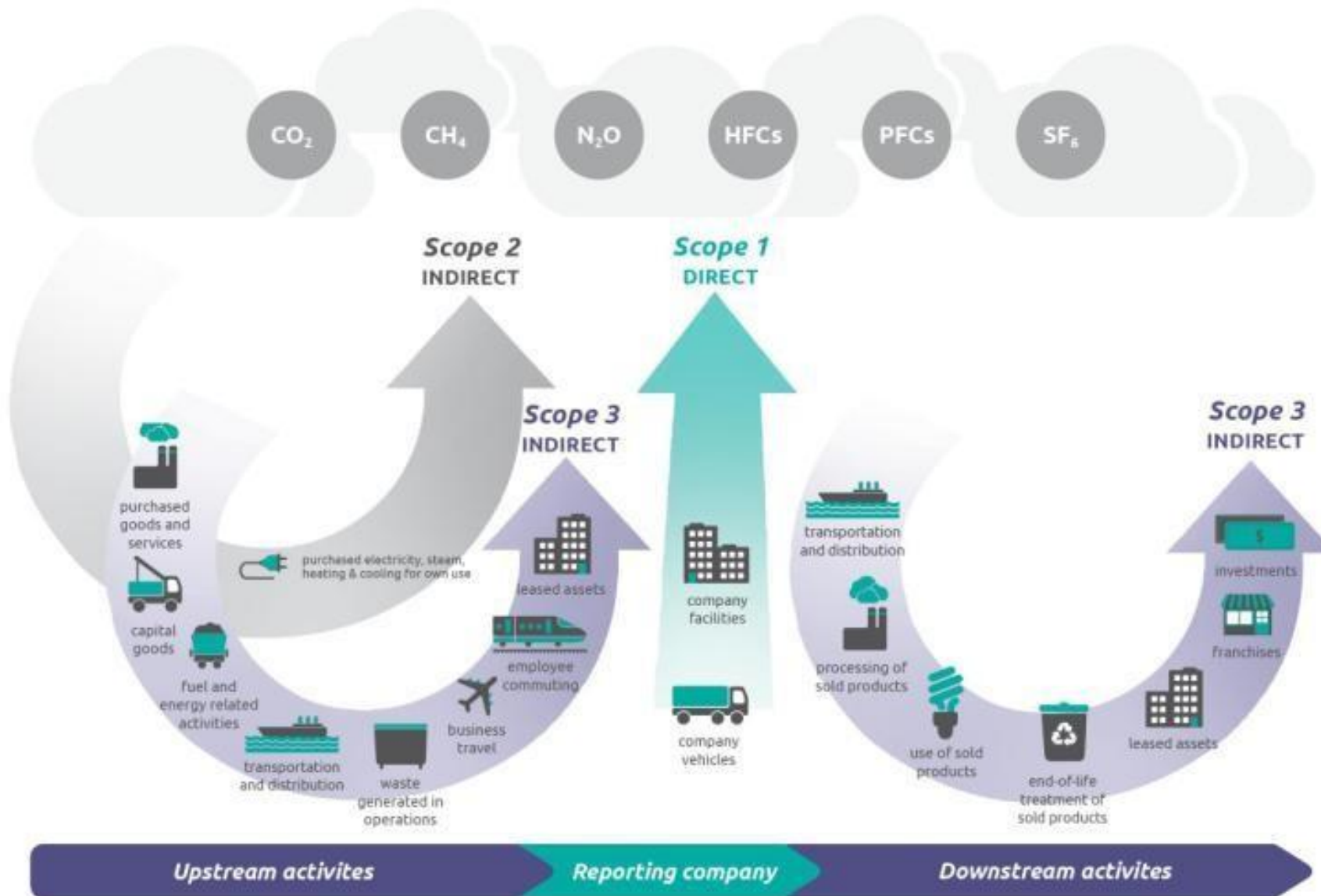
- *In-use emissions from running the building / asset;*

End user Carbon

Downstream



Figure [1.1] Overview of GHG Protocol scopes and emissions across the value chain



- **Direct emissions** are emissions from sources that are owned or controlled by the reporting company
- **Indirect emissions** are emissions that are a consequence of the activities of the company but occur at sources owned or controlled by another company

How to get a carbon footprint

- **A carbon footprint = activity data** (*e.g. litres of fuel used*) **x an emissions conversion factor** (*e.g. GHGs emitted per litre burnt*)
- **KgCO₂e (“equivalent”)** takes into account all the main GHGs emitted: CO₂, CH₄ and N₂O
- Think about **units of measurement** and converting between them: factors of a thousand
- Don’t forget you can calculate carbon from **proxy data** such as spend

<https://www.carbonfootprint.com/calculator.ASPX>

Assignment 5 – Calculate the carbon footprint of your own household